# JADEN SINGH

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## EDUCATION

**Bachelor's in Computer Science**, Carnegie Mellon University — Expected June 2027, 3.80 GPA
**Honors**: Dean's List (5 semesters)
**Current Research**: Member, GEM-AI Critical Minerals Project, U.S. Department of Energy Genesis Mission
**Technical Coursework**: Data Structures and Algorithms, Computer Systems, Parallel and Sequential Algorithms, Machine Learning, Probability and Computing, Functional Programming, Computation Theory

## TECHNICAL SKILLS

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| **Languages** | Python, R, Standard ML (coursework) |
| **AI, Data & ML** | pandas, NumPy, scikit-learn, XGBoost, River; time-series analysis, anomaly detection, feature engineering |
| **Systems & Geospatial** | Linux/Unix, Git, Docker, AWS S3, xarray, NetCDF, GeoPandas, GDAL/Rasterio, Parquet, Google Earth Engine, Streamlit |

## RESEARCH SOFTWARE

**CryoTartans - HyP3 Glacier Melt Processing** — Carnegie Mellon University

- Engineered a containerized Python CLI that acquires and reprojects Sentinel-1 rasters, rasterizes glacier masks, and assembles 2017-2024 NetCDF/xarray datacubes with optional AWS S3 output.
- Designed a tiled, parallelizable per-pixel time-series framework for melt extents and snowline elevations; validated monthly onset maps at $R^2 > 0.936$ against the reference method from May through August.

## WORK EXPERIENCE

**Machine Learning Intern** — Jul 2026 - Aug 2026
Gradiant — *Woburn, MA*

- Designed and deployed a Python/Streamlit predictive-maintenance application backed by cached Parquet geospatial processing; scored 54K+ water-main records for three-year failure risk at 6x the precision of industry heuristics among the highest-risk assets.
- Built a modular anomaly-detection pipeline for 10,224 readings from 29 meters; engineered 83 causal features and eight synthetic failure scenarios, benchmarked six batch and streaming models, and achieved 0.84 F1 on constant leaks and 0.81 F1 on intermittent leaks.
- Implemented time-based train/test splits and budget-aware top-k evaluation to validate models against operational heuristics while preventing temporal leakage.

**Wilderness Construction Crew Member** — Summers 2022-2025
Grand Teton National Park — *Moose, WY*

- Collaborated on remote bridge and historic-structure rebuilds while safely operating chainsaws, explosives, and mechanical-advantage rigging systems.

## SELECTED PROJECTS

**Teton Terrain Classifier** Built a Google Earth Engine Random Forest pipeline combining Sentinel-1/2 imagery and terrain features; achieved 96.1% validation accuracy against Dynamic World.

**Digital Altimeter** Designed and built the circuit, enclosure, and software for a barometric digital altimeter; interfaced a pressure sensor over I2C and implemented adjustable calibration with a real-time elevation display.